

BIOL-1 Practice Test 01

Modules 1-4: Study of Life through Cells

Instructions: This practice test covers material from Modules 1-4. Answer all questions to the best of your ability.

✂ TEAR-OFF ANSWER SHEET **Write your name and date above. Turn in this page when instructed.**

Multiple-choice answers 1. _____ 2. _____ 3. _____ 4. _____
5. _____ 6. _____ 7. _____ 8. _____ 9. _____ 10. _____

11. _____ 12. _____ 13. _____ 14. _____ 15. _____
16. _____ 17. _____ 18. _____ 19. _____ 20. _____
21. _____ 22. _____ 23. _____ 24. _____ 25. _____

Fill-in answers 1. _____

_____ 2.
_____ 3.
_____ 4.
_____ 5.
_____ 6.
_____ 7.
_____ 8.
_____ 9.
_____ 10.

One answer per line; use the question numbers printed in the booklet.

QUESTION BOOKLET — FREE RESPONSE **Answer the free-response questions on the lined space provided.** ## Part C: Short Answer (4 questions) *Answer each question in 2-3 complete sentences.* **36.** Name three characteristics that all living things share and briefly describe each one.

37. Explain the difference between an ionic bond and a covalent bond.

38. What are the four major types of biological macromolecules? Name the monomer (building block) for proteins.

39. Compare prokaryotic and eukaryotic cells. Give one example of each type of organism.

--- *End of Practice Test 01*

QUESTION BOOKLET — MULTIPLE CHOICE AND FILL-IN **Question booklet:

answer the questions below using the tear-off sheet for Parts A and B.** ## Part A: Multiple

Choice (25 questions) *Choose the best answer for each question.* ### Module 1: Study of

Life **1.** Which of the following is a characteristic shared by ALL living organisms? A)

Uses photosynthesis for energy B) Movement from place to place C) Made of one or more

cells D) Reproduces sexually **2.** What is the correct order of the levels of biological

organization, from smallest to largest? A) Cell → Tissue → Organ → Organ System →

Organism B) Organ → Tissue → Cell → Organ System → Organism C) Tissue → Cell →

Organ → Organism → Organ System D) Cell → Organ → Tissue → Organ System →

Organism **3.** In the scientific method, what comes AFTER making an observation? A)

Drawing a conclusion B) Forming a hypothesis C) Conducting an experiment D) Publishing

results **4.** In a controlled experiment, the variable that the scientist changes is called the:

A) Independent variable B) Constant C) Control variable D) Dependent variable **5.**

Homeostasis refers to: A) The ability to grow and develop B) Reproducing to create offspring

C) Maintaining a stable internal environment D) Responding to external stimuli **6.** An

organism that makes its own food through photosynthesis is called a(n): A) Consumer B)

Decomposer C) Autotroph D) Heterotroph --- ### Module 2: Basic Chemistry **7.** What

determines the atomic number of an element? A) Number of protons B) Number of neutrons

C) Atomic mass D) Number of electrons **8.** Isotopes of an element differ in their number

of: A) Electrons B) Protons C) Neutrons D) Electron shells **9.** When electrons are

TRANSFERRED between atoms, what type of bond is formed? A) Covalent bond B) Ionic

bond C) Hydrogen bond D) Polar bond **10.** Water is considered a polar molecule because:

A) It has a high specific heat B) It can dissolve many substances C) Electrons are shared

unequally between atoms D) It contains hydrogen atoms **11.** A solution with a pH of 2 is:

A) Neutral B) Strongly acidic C) Basic D) Weakly acidic **12.** A substance that resists

changes in pH is called a: A) Acid B) Buffer C) Base D) Solvent **13.** The tendency of

water molecules to stick together is called: A) Cohesion B) Surface tension C) Capillary action

D) Adhesion --- ### Module 3: Organic Molecules **14.** Which element is the "backbone"

of all organic molecules? A) Nitrogen B) Oxygen C) Hydrogen D) Carbon **15.** What is

the relationship between monomers and polymers? A) Polymers are the building blocks of

monomers B) Monomers break down polymers C) Monomers link together to form polymers

D) They are the same thing **16.** In dehydration synthesis, polymers are built by: A)

Adding energy without changing water B) Removing water molecules C) Adding water

molecules D) Breaking bonds **17.** Which biomolecule is the body's primary source of quick energy? A) Protein B) Nucleic acid C) Lipid D) Carbohydrate **18.** What type of bond holds amino acids together in a protein? A) Glycosidic bond B) Hydrogen bond C) Ionic bond D) Peptide bond **19.** When a protein loses its shape due to heat or pH changes, this is called: A) Denaturation B) Polymerization C) Hydrolysis D) Condensation --- ### Module 4: Cells **20.** According to cell theory, all cells: A) Contain a nucleus B) Come from pre-existing cells C) Are identical in structure D) Can survive independently **21.** The main difference between prokaryotic and eukaryotic cells is that prokaryotes: A) Are larger than eukaryotes B) Have more organelles C) Cannot reproduce D) Lack a membrane-bound nucleus **22.** Which organelle is called the "powerhouse of the cell" because it produces ATP? A) Mitochondria B) Golgi apparatus C) Nucleus D) Ribosome **23.** Which structure is found in plant cells but NOT in animal cells? A) Mitochondria B) Ribosomes C) Nucleus D) Cell wall **24.** Proteins are synthesized (made) at which organelle? A) Ribosome B) Mitochondria C) Lysosome D) Golgi apparatus **25.** The Golgi apparatus functions to: A) Digest waste materials B) Produce ATP C) Store DNA D) Package and ship proteins --- ## Part B: Fill in the Blank (10 questions) *Write the correct term in the blank.* **26.** The basic unit of life is the _____. **27.** The _____ is the organelle that contains the cell's genetic material (DNA). **28.** Atoms of the same element with different numbers of neutrons are called _____. **29.** The process of breaking down polymers by adding water is called _____. **30.** The four most common elements in living things are carbon, hydrogen, oxygen, and _____. **31.** A solution with a pH less than 7 is _____. **32.** The monomer (building block) of proteins is the _____. **33.** Organisms that cannot make their own food and must consume other organisms are called _____. **34.** The _____ is the membrane system that includes the ER and Golgi apparatus. **35.** According to _____ theory, mitochondria and chloroplasts were once free-living bacteria. ---